

A strict lower bound on a norm's unit sphere

MathlibAnnex.FiniteSup.FinMapAlmostIsometric

definition

Defines the strict lower condition on a fixed norm unit sphere.

Statement

Let $n, N \in \mathbb{N}$ be arbitrary, including zero. Use reference coordinates $V = \mathbb{R}^n$ with the sup norm and standard basis. The supplied continuous seminorm p has constants $a_p, b_p > 0$ satisfying

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm; it can differ from the reference sup norm. Fix any $\varepsilon \in \mathbb{R}$ and continuous real linear $A : V \rightarrow \mathbb{R}^N$, with the target sup norm. Define the property by

$$\forall x \in V, \quad p(x) = 1 \implies 1 - \varepsilon < \|Ax\|_\infty.$$

Definition

The property requires the displayed strict lower bound at every p -unit vector. It is a condition on the given map A , not a construction of a new map.

Assumptions

The inputs are the continuous seminorm p with its positive two-sided comparison bounds, any real ε , and the fixed continuous real linear map A . Neither an upper estimate on A , positivity of ε , nor injectivity of A is an input to this definition.

Conclusion

This is only the lower unit-sphere condition. In dimension zero the unit sphere is empty and the condition is vacuous.

As a separate consequence, let $x \neq 0$. Positive definiteness gives $p(x) > 0$, and

$$p(x/p(x)) = 1, \quad \|A(x/p(x))\|_\infty = \|Ax\|_\infty / p(x).$$

Apply the condition at that same vector and multiply by $p(x) > 0$ to obtain

$$(1 - \varepsilon)p(x) < \|Ax\|_\infty.$$

This is the lower bound away from zero. It adds no upper estimate.

Main citations

- Exact declaration and source
- the lower bound away from zero

Lean source signature (exact)

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def FinMapAlmostIsometric {n N : ℕ} (M : NormModel n) (ε : ℝ)
  (A : Coord n →L[ℝ] SupCoord N) : Prop :=
  ∀ x : Coord n, M.p x = 1 → 1 - ε < ‖A x‖
```

In the source	Mathematical meaning
Coord n; SupCoord N; →L[ℝ]	$\mathbb{R}^n, \mathbb{R}^N$ with sup norms; private aliases mean $\text{Fin } n \rightarrow \mathbb{R}$ and $\text{Fin } N \rightarrow \mathbb{R}$. The arrow is continuous real linear.
M : NormModel n; M.p	The continuous norm p with positive comparisons $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$.
ε : ℝ; A : Coord n →L[ℝ] SupCoord N	An arbitrary real ε and the supplied map $A : \mathbb{R}^n \rightarrow \mathbb{R}^N$.

In the source	Mathematical meaning
FinMapAlmostIsometric M ε A : Prop	A proposition about this same p, ε, A , defined by the entire RHS below.
∀ x : Coord n, M.p x = 1 → 1 - ε < ‖A x‖	For every $x \in V$, if $p(x) = 1$ then $1 - \varepsilon < \ Ax\ _\infty$. $\forall$ is universal quantification and $\rightarrow$ is implication. Contractivity is absent.

The braces in $\{n \ N : \mathbb{N}\}$ let Lean infer the nonnegative dimensions. The final $:=$ gives the defining condition; the earlier arrows in the type of A describe a linear map, whereas the arrow after $M.p \ x = 1$ is logical implication.

Surrounding assumptions and aliases.

▼ Exact content identity

Declaration: MathlibAnnex.FiniteSup.FinMapAlmostIsometric

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